A Rigorous Galerkin Closure for the Elastodynamic Volume Integral Equation on a Cube

27-Component Path-B Formulation

Research Notes

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1 Motivation: differentiating under the integral fails

The classical moment-method closure for the cube T-matrix begins from the Lippmann–Schwinger (LS) equation

$$u_i(\boldsymbol{r}) \;=\; u_i^0(\boldsymbol{r}) + \omega^2 \int_V G_{ij}(\boldsymbol{r}-\boldsymbol{r}')\,\delta\rho(\boldsymbol{r}')\,u_j(\boldsymbol{r}')\,dV' - \int_V \partial'_k G_{in}(\boldsymbol{r}-\boldsymbol{r}')\,\delta c_{nklj}(\boldsymbol{r}')\,\partial'_l u_j(\boldsymbol{r}')\,dV', \quad (1)$$

where G_{ij} is the elastostatic Green's tensor and $V \subset \mathbb{R}^3$ is the inclusion. To obtain a closed

finite-dimensional system one then formally takes derivatives at the centre $\boldsymbol{r} = \mathbf{0}$:

$$
\begin{aligned}
u_i(\mathbf{0}) &= u_i^0(\mathbf{0}) + \omega^2 \int_V G_{ij}\, \delta\rho\, u_j\, dV' - \int_V \partial'_k G_{in}\, \delta c_{nklj}\, \partial'_l u_j\, dV', \\
\partial_p u_i(\mathbf{0}) &= \partial_p u_i^0(\mathbf{0}) - \omega^2 \int_V \partial'_p G_{ij}\, \delta\rho\, u_j\, dV' + \int_V \partial'_p \partial'_k G_{in}\, \delta c_{nklj}\, \partial'_l u_j\, dV', \\
\partial_q \partial_p u_i(\mathbf{0}) &= \partial_q \partial_p u_i^0(\mathbf{0}) + \omega^2 \int_V \partial'_q \partial'_p G_{ij}\, \delta\rho\, u_j\, dV' - \int_V \partial'_q \partial'_p \partial'_k G_{in}\, \delta c_{nklj}\, \partial'_l u_j\, dV',
\end{aligned}
\tag{2}
$$

where the r–derivatives have been converted to r'–derivatives by translation invariance $\partial_r G(\boldsymbol{r} - \boldsymbol{r}') = -\partial_{r'} G(\boldsymbol{r} - \boldsymbol{r}')$.

The Leibniz rule used in passing from (1) to (2) requires (DCT version) a dominating function $|\partial_r f(\boldsymbol{r}, \boldsymbol{r}')| \le g(\boldsymbol{r}')$ with $g \in L^1(V)$. The static elastic kernel has the singularity

$$
G_{ij}^{\mathrm{stat}}(\boldsymbol{s}) \sim \frac{1}{|\boldsymbol{s}|}, \quad \partial' G \sim \frac{1}{|\boldsymbol{s}|^2}, \quad \partial'\partial' G \sim \frac{1}{|\boldsymbol{s}|^3}, \quad \partial'\partial'\partial' G \sim \frac{1}{|\boldsymbol{s}|^4}, \tag{3}
$$

so already at the level of $\partial' G$ *no integrable dominating function exists* when $\boldsymbol{r}$ is in the support of the inclusion. Each of the equations in (2) therefore makes only *formal* sense; converting them into rigorous statements requires distributional regularization (Cauchy principal value, Hadamard finite part, and Eshelby-type delta corrections).

LS derivative order	worst kernel	status
0 (body force)	$1/\lvert\boldsymbol{s}\rvert$	L^1_{loc} ✓
0 (elastic kernel)	$1/\lvert\boldsymbol{s}\rvert^2$	conditionally convergent (PV)
1 (body force)	$1/\lvert\boldsymbol{s}\rvert^2$	PV
1 (elastic kernel)	$1/\lvert\boldsymbol{s}\rvert^3$	**Eshelby delta required**
2 (body force)	$1/\lvert\boldsymbol{s}\rvert^3$	**Eshelby delta required**
2 (elastic kernel)	$1/\lvert\boldsymbol{s}\rvert^4$	**derivative-of-delta / Hadamard PF**

Table 1: Singular structure of the formal closure equations (2).

2 Path B: Galerkin formulation on a polynomial trial space

We replace point evaluation by a Galerkin (weak) projection onto a finite-dimensional polynomial trial space. The resulting bilinear forms involve only G_{ij} itself (the singularity $1/|\boldsymbol{s}|$ is locally L^1 in $\mathbb{R}^3$), so *every entry of the closure matrix is an absolutely convergent integral.* No principal values, no delta corrections, no distributional bookkeeping.

2.1 Trial space T_{27}

Let $V = [-a, a]^3$. Denote by $P_2(V)$ the space of real polynomials in $\boldsymbol{r} = (r_1, r_2, r_3)$ of total degree ≤ 2, $\dim P_2(V) = 10$. The naive vector trial space $(P_2(V))^3$ has dimension 30. We remove the 3 rigid-rotation modes by symmetrizing the linear part, giving the 27-dimensional subspace

$$
T_{27} \;=\; \Big\{ \boldsymbol{\varphi}(\boldsymbol{r}) \;\Big|\; \varphi_i(\boldsymbol{r}) = U_i + \varepsilon_{ai}\, r_a + \tfrac{1}{2}\, M_{abi}\, r_a r_b, \quad \varepsilon_{ai} = \varepsilon_{ia}, \quad M_{abi} = M_{bai} \Big\}. \tag{4}
$$

The dimensions add as $3 + 6 + 18 = 27$, matching one displacement monopole, the symmetric strain tensor, and the symmetric second-derivative tensor of $\boldsymbol{u}$ (equivalent to the strain gradient $\partial_k \varepsilon_{ij}$ through the kinematic identity $\partial_k \partial_l u_m = \partial_k \varepsilon_{lm} + \partial_l \varepsilon_{km} - \partial_m \varepsilon_{kl}$).

2.2 Galerkin LS

Definition 1 (Path-B closure)**.** The Path-B 27-component closure is the variational problem: find $\boldsymbol{u}^h \in T_{27}$ such that for all $\boldsymbol{\varphi} \in T_{27}$,

$$\langle \boldsymbol{\varphi}, \boldsymbol{u}^h \rangle_V \;=\; \langle \boldsymbol{\varphi}, \boldsymbol{u}^0 \rangle_V + \omega^2 \delta\rho\, \mathcal{B}_{\text{body}}[\boldsymbol{\varphi}, \boldsymbol{u}^h] - \delta c_{nklj}\, \mathcal{B}_{\text{el}}[\boldsymbol{\varphi}, \boldsymbol{u}^h], \tag{5}$$

where the bilinear forms are

$$\langle \boldsymbol{\varphi}, \boldsymbol{u} \rangle_V := \int_V \varphi_i(\boldsymbol{r})\, u_i(\boldsymbol{r})\, dV, \tag{6}$$

$$\mathcal{B}_{\text{body}}[\boldsymbol{\varphi}, \boldsymbol{u}] := \int_V\!\!\int_V \varphi_i(\boldsymbol{r})\, G_{in}(\boldsymbol{r} - \boldsymbol{r}')\, u_n(\boldsymbol{r}')\, dV\, dV', \tag{7}$$

$$\mathcal{B}_{\text{el}}[\boldsymbol{\varphi}, \boldsymbol{u}] := \int_V\!\!\int_V \varphi_i(\boldsymbol{r})\, \partial'_k G_{in}(\boldsymbol{r} - \boldsymbol{r}')\, \partial'_l u_j(\boldsymbol{r}')\, dV\, dV'. \tag{8}$$

2.3 Why this is well defined

Proposition 2. *For polynomial $\boldsymbol{\varphi}, \boldsymbol{u} \in T_{27}$, every bilinear form in (5) is an absolutely convergent integral. No distributional regularization is needed.*

Proof sketch. The mass form is trivial because the integrand is a polynomial. For $\mathcal{B}_{\text{el}}$, we use integration by parts on $\boldsymbol{u}$ (legitimate because $\boldsymbol{u}$ is a polynomial and the boundary term is a 2D integral against a smooth L^1_{loc} kernel):

$$\mathcal{B}_{\text{el}}[\boldsymbol{\varphi}, \boldsymbol{u}] = -\int_V\!\!\int_V \varphi_i\, G_{in}\, \partial'_k \partial'_l u_j\, dV'\, dV + \int_V \!\oint_{\partial V} \varphi_i\, G_{in}\, \partial'_l u_j\, n_k\, dS'\, dV. \tag{9}$$

Both integrands now contain only $G_{in}(\boldsymbol{r} - \boldsymbol{r}') \sim 1/|\boldsymbol{r} - \boldsymbol{r}'|$ multiplied by polynomials/smooth functions. The 6-fold integral $\iint_{V\times V} 1/|\boldsymbol{r} - \boldsymbol{r}'|\, dV\, dV'$ converges by the Hardy–Littlewood–Sobolev inequality (or directly: in $\mathbb{R}^6$ the singular set $\{\boldsymbol{r} = \boldsymbol{r}'\}$ has codimension 3 and $1/|\boldsymbol{r} - \boldsymbol{r}'|$ has order-1 singularity, hence is locally L^1 in $\mathbb{R}^6$). The 5-fold boundary integral is similar. The body form is even simpler since no integration by parts is needed.

□ □

2.4 Equivalence to point-collocation in the limit

The classical Path-A closure (point evaluation at $\boldsymbol{r} = \boldsymbol{0}$) is the formal limit of a Petrov–Galerkin scheme with Dirac-delta test functions. Both Path A and Path B converge to the exact volume integral equation as the polynomial degree $\to \infty$; at finite truncation their stiffness matrices differ but encode the same physical operator. Path B is preferable for analysis because it is symmetric, energy conserving, and free of singular regularizations.

3 Center-of-mass / relative coordinates

The 6-fold integrals in (5) are reduced to 3-fold integrals against the *cube autocorrelation* via the *centre-of-mass and relative coordinate* change of variables (also called *sum-and-difference variables* in the Eshelby/Mura literature, or *Jacobi coordinates* in the simplest 2-body case):

$$\boldsymbol{\xi} = \tfrac{1}{2}(\boldsymbol{r} + \boldsymbol{r}'), \qquad \boldsymbol{s} = \boldsymbol{r} - \boldsymbol{r}', \qquad \boldsymbol{r} = \boldsymbol{\xi} + \tfrac{\boldsymbol{s}}{2}, \quad \boldsymbol{r}' = \boldsymbol{\xi} - \tfrac{\boldsymbol{s}}{2}, \qquad d\boldsymbol{r}\, d\boldsymbol{r}' = d\boldsymbol{\xi}\, d\boldsymbol{s}. \tag{10}$$

The constraint $\boldsymbol{r}, \boldsymbol{r}' \in V$ becomes $\boldsymbol{\xi} \in V \cap (V + \boldsymbol{s}/2) \cap (V - \boldsymbol{s}/2)$, which for the cube $V = [-a, a]^3$ simplifies to a centred box of half-lengths $b_i(\boldsymbol{s}) = a - |s_i|/2$:

$$\boldsymbol{\xi} \in B(\boldsymbol{s}) := \prod_{i=1}^{3} \big[-(a - |s_i|/2),\, a - |s_i|/2 \big], \qquad \text{valid when } |s_i| < 2a. \tag{11}$$

The 6-fold integral becomes

$$\int_V\!\!\int_V P(\boldsymbol{r})\,K(\boldsymbol{r}-\boldsymbol{r}')\,Q(\boldsymbol{r}')\,d\boldsymbol{r}\,d\boldsymbol{r}' \;=\; \int_{[-2a,2a]^3} K(\boldsymbol{s})\,\underbrace{\Big[\int_{B(\boldsymbol{s})} P\big(\boldsymbol{\xi}+\tfrac{\boldsymbol{s}}{2}\big)\,Q\big(\boldsymbol{\xi}-\tfrac{\boldsymbol{s}}{2}\big)\,d\boldsymbol{\xi}\Big]}_{\text{polynomial in }(\boldsymbol{s},a)}\,d\boldsymbol{s}. \tag{12}$$

For polynomial P, Q the inner integral is elementary: every ξ_i^p integrates over $[-b_i, b_i]$ to $\frac{2b_i^{p+1}}{p+1}$ when p is even and to 0 when p is odd, giving a polynomial in $(b_1, b_2, b_3) = (a - |s_1|/2,\ a - |s_2|/2,\ a - |s_3|/2)$.

Cube autocorrelation. For $P = Q = 1$ the inner integral is exactly the autocorrelation of the cube indicator,

$$(\chi_V \star \chi_V)(\boldsymbol{s}) \;=\; \prod_{i=1}^{3} \big(2a - |s_i|\big)_+, \tag{13}$$

the 3-D *tent function* $\Lambda_V(\boldsymbol{s})$.

4 Master integrals

After the centre-of-mass/relative reduction, every entry of the 27×27 Galerkin matrix is a finite linear combination of the following *master integrals* over the doubled cube $\Omega = [-2a, 2a]^3$:

$$\mathcal{I}_{p,q,r}(a) := \int_\Omega \frac{s_1^p\, s_2^q\, s_3^r}{|\boldsymbol{s}|}\,d\boldsymbol{s}, \tag{14}$$

$$\mathcal{J}^{(ij)}_{p,q,r}(a) := \int_\Omega \frac{s_i s_j\, s_1^p\, s_2^q\, s_3^r}{|\boldsymbol{s}|^3}\,d\boldsymbol{s}. \tag{15}$$

For the 9-component (current) closure only $p + q + r \le 3$ is needed; the 27-component closure requires $p + q + r \le 5$.

By the parity $\boldsymbol{s} \to -\boldsymbol{s}$ on each axis separately, $\mathcal{I}_{p,q,r}$ vanishes whenever any of p, q, r is odd, and $\mathcal{J}^{(ij)}_{p,q,r}$ vanishes unless the multiset $\{i, j, p\text{-copies of } 1, q\text{-copies of } 2, r\text{-copies of } 3\}$ has even count for each axis. This collapses the formally $\binom{8}{3} \cdot 3 = 168$ entries down to ~ 20 inequivalent integrals after cubic symmetry.

4.1 Tier-1 results (computed analytically)

The basic integral $\mathcal{I}_{0,0,0}(a)$ is the dimensionless cube constant g_0 scaled by a^2. Mathematica's `Integrate` returns it in the form

$$\int_{[-1,1]^3} \frac{1}{|\boldsymbol{u}|}\,du \;=\; -2\Big(\pi + \log\big(1351 - 780\sqrt{3}\big)\Big), \tag{16}$$

which equals the Pell-stable form used in the existing `CubeAnalytic.wl` package via the identity $1351 - 780\sqrt{3} = (70226 + 40545\sqrt{3})^{-2/3}$ (both sides are Pell units of $\mathbb{Z}[\sqrt{3}]$, satisfying $1351^2 - 3 \cdot 780^2 = 1$ and $70226^2 - 3 \cdot 40545^2 = 1$). Numerical agreement to $\sim$78 digits has been verified.

4.2 Verified analytical master integrals

The verification at degree 0 was reported in Section 4.1. We extend the table to include all parity-allowed master integrals up to degree 6:

The same family of integrals over the positive octant $[0, 1]^3$ supplies the body-bilinear-form building blocks; we collect them in Table 3.

(p,q,r)	$\mathcal{M}_{p,q,r} = \int_{[-1,1]^3} \frac{x^p y^q z^r}{\lvert \boldsymbol{r} \rvert}\, dV$	numerical
$(0,0,0)$	$-2\pi - \frac{4}{3}\log(70226 - 40545\sqrt{3})$	9.5203095
$(2,0,0)$	$\frac{1}{9}\big(6\sqrt{3} - \pi + \log(3650401 + 2107560\sqrt{3})\big)$	2.5615786
$(2,2,0)$	$\frac{1}{135}\big(108\sqrt{3} + 18\pi + 63\log 2 + 20\log(26 - 15\sqrt{3})$	0.75095335
	$- 60\log(\sqrt{3}-1) - 66\log(1+\sqrt{3}) - 90\coth^{-1}\sqrt{3}\big)$	
$(4,0,0)$	$-\frac{2}{45}\big(12\sqrt{3} + 7\pi - 42\log 2$	1.4351485
	$+ 99\log(\sqrt{3}-1) - 15\log(1+\sqrt{3})\big)$	
$(2,2,2)$	$\frac{1}{1260}\big(1008\sqrt{3} - 42\pi + 522\log 2 + 441\log(\sqrt{3}-1)$	0.22735457
	$- 1395\log(1+\sqrt{3}) - 84\log(2+\sqrt{3}) - 90\log(11\sqrt{3}-19)$	
	$+ 35\log(26+15\sqrt{3}) - 672\coth^{-1}\sqrt{3}\big)$	
$(4,2,0)$	$\frac{1}{630}\big(-48\sqrt{3} + 17\pi$	0.42942069
	$+ 12\tanh^{-1}\frac{21252634831\sqrt{3}}{36810643322}\big)$	
$(6,0,0)$	$\frac{1}{42}\big(-5\pi - 4\log 2 + 8(3\sqrt{3} + \log(5+3\sqrt{3}))\big)$	0.99201788

Table 2: Closed-form values of the scalar master integrals $\mathcal{M}_{p,q,r}$ over the unit cube. Cubic symmetry equates entries that are permutations of each other; only canonical orderings $p \geq q \geq r$ are listed.

4.3 Body bilinear-form building blocks

For the constant subspace of T_{27} the diagonal and off-diagonal entries of the body bilinear form $\mathcal{B}_{\text{body}}$ reduce, after the centre-of-mass / relative-coordinate substitution and an octant-symmetry restriction, to three positive scalars. Define

$$i_A := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\, \frac{1}{|\boldsymbol{u}|}\, d\boldsymbol{u}, \tag{17}$$

$$i_{B,11} := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\, \frac{u_1^2}{|\boldsymbol{u}|^3}\, d\boldsymbol{u}, \tag{18}$$

$$i_{B,12} := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\, \frac{u_1 u_2}{|\boldsymbol{u}|^3}\, d\boldsymbol{u}. \tag{19}$$

The first reduces by linearity to a signed combination of $\mathcal{M}^+_{p,q,r}$ values (Table 3):

$$i_A \;=\; \mathcal{M}^+_{0,0,0} \;-\; 3\mathcal{M}^+_{1,0,0} \;+\; 3\mathcal{M}^+_{1,1,0} \;-\; \mathcal{M}^+_{1,1,1} \;\approx\; 0.2352890805487075. \tag{20}$$

The remaining two are obtained by integrating once by parts in the u_1 direction; the singular kernel $|\boldsymbol{u}|^{-3}$ collapses to a $|\boldsymbol{u}|^{-1}$-class kernel and a clean two-dimensional residual:

$$i_{B,11} \;=\; i_A \;-\; J_{11}, \qquad J_{11} \;=\; \int_{[0,1]^2} (1-u_2)(1-u_3)\Big[\sqrt{1+u_2^2+u_3^2} - \sqrt{u_2^2+u_3^2}\Big]\, du_2\, du_3, \tag{21}$$

$$i_{B,12} \;=\; \int_{[0,1]^2} (1-u_2)(1-u_3)\, u_2 \Big[\frac{1}{\sqrt{u_2^2+u_3^2}} \;-\; \operatorname{arcsinh}\Big(\frac{1}{\sqrt{u_2^2+u_3^2}}\Big)\Big]\, du_2\, du_3. \tag{22}$$

Mathematica evaluates both two-dimensional integrals in closed form:

$$J_{11} \;=\; \frac{1}{240}\Big(8 + 8\sqrt{2} - 16\sqrt{3} - 55\log(2) + 15\log\left(3+2\sqrt{2}\right) + 110\log\left(1+\sqrt{3}\right) - 15\log\left(2+\sqrt{3}\right) - 80\tan^{-1}\left(\frac{3}{8}\right) + 80\tan^{-1}\left(\frac{1}{183}\left(96 - 73\sqrt{3}\right)\right) + 10\sinh^{-1}(1)\Big)$$

with $J_{11} \approx 0.1568593870324717$, and

$$i_{B,12} \;=\; \frac{1}{240}\Big(-8\sqrt{2} + 4\sqrt{3} + 20\log(2) + 5\log(256) + 40\log\left(1+\sqrt{2}\right) + 22\log\left(3+2\sqrt{2}\right) - 120\log\left(1+\sqrt{3}\right) - 22\log\left(2+\sqrt{3}\right) + 20\tan^{-1}\left(\frac{1}{2}\right) + 16\tan^{-1}\left(\frac{5}{8}\right) - 40\tan^{-1}\left(\frac{3}{2}\right) + 40\tan^{-1}\left(8+\frac{13}{\sqrt{3}}\right) - 20\tan^{-1}\left(\frac{1}{11}\left(8-5\sqrt{3}\right)\right) - 16\tan^{-1}\left(\frac{1}{167}\left(160+89\sqrt{3}\right)\right) + 4\sinh^{-1}(1) + 36\sinh^{-1}\left(\frac{1}{\sqrt{2}}\right)\Big)$$

(p,q,r)	$\mathcal{M}^{+}_{p,q,r} = \int_{[0,1]^3} \frac{x^p y^q z^r}{\lvert \boldsymbol{r} \rvert}\, dV$	numerical
$(0,0,0)$	$-\frac{\pi}{4} - \frac{1}{6}\log\big(70226 - 40545\sqrt{3}\big)$	1.1900387
$(1,0,0)$	$\frac{1}{36}\big(-2\pi - 12\big(\sqrt{2} - \sqrt{3} + \sinh^{-1}(1)\big)$ $+ \cosh^{-1}(189750626) + 18\coth^{-1}\sqrt{3}\big)$	0.51559356
$(1,1,0)$	$\frac{1}{12}\big(1 - 7\sqrt{2} + 6\sqrt{3} + \log(5\sqrt{2} - 7)$ $- 3\log(2 - \sqrt{3})\big)$	0.23329690
$(2,0,0)$	$\frac{1}{144}\big(12\sqrt{3} - 2\pi$ $+ \log(26650854921601 + 15386878263120\sqrt{3})\big)$	0.32019732
$(1,1,1)$	$\frac{1}{5}\big(1 - 4\sqrt{2} + 3\sqrt{3}\big)$	0.10785963
$(2,1,0)$	$\frac{1}{2880}\big(-504\sqrt{2} + 672\sqrt{3} + 32\pi + 72\log(\sqrt{3} - 1)$ $- 252\log(1 + \sqrt{3}) + 90\log(2(2 + \sqrt{3})) - 261\sinh^{-1}(1)$ $+ 40\cosh^{-1}(26) + 45\coth^{-1}\sqrt{2}\big)$	0.14741068

Table 3: Closed-form values of the positive-octant master integrals $\mathcal{M}^{+}_{p,q,r}$ over $[0,1]^3$. Up to canonical ordering $p \geq q \geq r$ and $p + q + r \leq 3$ they exhaust all polynomial weights needed by the Path-B body bilinear form on the constant-and-linear subspace.

with $i_{B,12} \approx 0.04970171296728090$.

A non-trivial consistency check is supplied by the trace identity

$$3\, i_{B,11} \;=\; \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\, \frac{u_1^2 + u_2^2 + u_3^2}{|\boldsymbol{u}|^3}\, d\boldsymbol{u} \;=\; i_A,$$

which is verified analytically and numerically to working precision.

4.4 The 9×9 body bilinear matrix on T_9

With the master integrals of Sections 4.2 and the three scalar building blocks i_A, $i_{B,11}$, $i_{B,12}$, the body bilinear form $\mathcal{B}_9 : T_9 \times T_9 \to \mathbb{R}$ is assembled entry-by-entry from

$$\mathcal{B}_9[i,j] \;=\; 32 \sum_{p,q=1}^{3} \Big\{ A\, \delta_{pq}\, K_1[\Pi_{ij,pq}] \;+\; B\, K_3[\Pi_{ij,pq};\, p,q] \Big\}, \tag{23}$$

where $\Pi_{ij,pq}(\boldsymbol{u})$ is the residual polynomial obtained by (a) substituting $\phi_p^{(i)}(\boldsymbol{r})\phi_q^{(j)}(\boldsymbol{r}')$, (b) integrating the centre-of-mass variable $\boldsymbol{\xi}$ over the overlap box, and (c) dividing out the standard tent factor $(1-u_1)(1-u_2)(1-u_3)$. The prefactor $32 = 4 \cdot 8/1$ absorbs the $s \mapsto 2u$ Jacobian 8, the Green-tensor rescaling $G(2\boldsymbol{u}) = \frac{1}{2}[A/|\boldsymbol{u}| + B\, u_p u_q/|\boldsymbol{u}|^3]$, and the symmetric-box factor 2/1. A further factor of 8 appears on parity-even entries when we restrict to the positive octant $[0,1]^3$, so a diagonal displacement-block entry is proportional to 256.

Under cubic symmetry the T_9-basis decomposes into four irreducible blocks, each of which is characterised by at most two independent scalars:

- **Displacement block** ($i,j \in \{1,2,3\}$): diagonal, with a single scalar.
- **Displacement–strain block** ($i \in \{1,2,3\}$, $j \in \{4,\ldots,9\}$): identically zero by parity.
- **Axial-strain block** ($i,j \in \{4,5,6\}$): diagonal entry α, off-diagonal entry β.
- **Axial–shear block** ($i \in \{4,5,6\}$, $j \in \{7,8,9\}$): identically zero by parity.
- **Shear-strain block** ($i,j \in \{7,8,9\}$): diagonal entry γ, off-diagonals identically zero by parity.

Displacement block

All three displacement-block diagonal entries are equal and reduce in closed form to

$$\mathcal{B}_9[1,1] \;=\; \mathcal{B}_9[2,2] \;=\; \mathcal{B}_9[3,3] \;=\; 256\,\big(A\,i_A \;+\; B\,i_{B,11}\big) \;=\; 60.23400462046913\,A \;+\; 20.07800154015638\,B, \tag{24}$$

with off-diagonal entries vanishing by parity.

Axial-strain block

The diagonal entry $\alpha := \mathcal{B}_9[4,4] = \mathcal{B}_9[5,5] = \mathcal{B}_9[6,6]$ splits into

$$\alpha \;=\; \tfrac{256}{3}\Big[A\big(i_A - 2K_1^{(2,0,0)}\big) \;+\; B\big(i_{B,11} - 2K_3^{(2,0,0)}\big)\Big] \;=\; 15.37252824403423\,A \;+\; 3.949761457945144\,B, \tag{25}$$

where

$$K_1^{(2,0,0)} \;:=\; \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\,\frac{u_1^2}{|\boldsymbol{u}|}\,d\boldsymbol{u}, \quad K_3^{(2,0,0)} \;:=\; \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\,\frac{u_1^4}{|\boldsymbol{u}|^3}\,d\boldsymbol{u}.$$

The off-diagonal entry $\beta := \mathcal{B}_9[4,5]$ is purely a B-channel correction,

$$\beta \;=\; -256\,B\,K_{3,\mathrm{off}}^{(1,1,0)} \;=\; 0\,A \;-\; 1.471925680511371\,B, \tag{26}$$

where $K_{3,\mathrm{off}}^{(1,1,0)} = \int_{[0,1]^3}(1-u_1)(1-u_2)(1-u_3)\,u_1u_2\,\frac{u_1u_2}{|\boldsymbol{u}|^3}\,d\boldsymbol{u}$. Note that β vanishes in the A-channel because the integrand of the A piece is odd in (u_1,u_2).

Shear-strain block

The diagonal entry $\gamma := \mathcal{B}_9[7,7] = \mathcal{B}_9[8,8] = \mathcal{B}_9[9,9]$ evaluates to

$$\gamma \;=\; \tfrac{128}{3}\Big[A\big(i_A - 2K_1^{(2,0,0)}\big) + B\big(i_{B,11} - 2K_3^{(0,2,0)}\big)\Big] \;-\; 128\,B\,K_{3,\mathrm{off}}^{(1,1,0)} \;=\; 7.686264122017116\,A \;+\; 2.119728856266587\,B, \tag{27}$$

where $K_3^{(0,2,0)} = \int_{[0,1]^3}(1-u_1)(1-u_2)(1-u_3)\,\frac{u_2^2u_1^2}{|\boldsymbol{u}|^3}\,d\boldsymbol{u}$. The relation $\gamma^{(A)} = \frac{1}{2}\,\alpha^{(A)}$ is an exact identity: the A-channel of γ is exactly half the A-channel of α, coming from the bilinearity of the two-component shear basis $\phi^{(7)} = (0, r_3/2, r_2/2)$.

Numerical verification of cubic symmetry (all six independent identities) was performed to 10^{-20} precision; see the Mathematica script output.

4.5 The 18×18 quad–quad block on T_{27}

Extending T_9 to T_{27} adjoins the 18 quadratic vector fields

$$\phi^{(9+6(k-1)+m)}(\boldsymbol{r}) \;=\; q_m(\boldsymbol{r})\,\boldsymbol{\varepsilon}_k, \qquad k \in \{1,2,3\},\;\; m \in \{1,\dots,6\},$$

where the six quadratic monomials in canonical Voigt order are

$$q_1 = r_1^2, \quad q_2 = r_2^2, \quad q_3 = r_3^2, \quad q_4 = r_2r_3, \quad q_5 = r_1r_3, \quad q_6 = r_1r_2.$$

We refer to $q_{1..3}$ as the squared (S) modes and to $q_{4..6}$ as the cross (X) modes. On the 18×18 quad–quad sub-block, the body bilinear form decomposes into an A-channel and a B-channel,

$$\mathcal{B}_{27}^{\mathrm{quad}}[i,j] \;=\; 32\,\big(a_{ij}\,A \;+\; b_{ij}\,B\big), \tag{28}$$

where the scalars a_{ij}, b_{ij} depend only on the orbit of (i,j) under the combined O_h action on the three coordinate axes and the $i \leftrightarrow j$ symmetry.

Enumerating all $18^2 = 324$ index pairs and partitioning by this action yields exactly **34 orbits**. Of these, **20 orbits** (total multiplicity 222) are identically zero in both channels: each such vanishing is a parity cancellation forced by cubic symmetry (the integrand is odd in at least one coordinate axis under the $[-1,1]^3$ integration). The remaining **14 orbits** (total multiplicity 102) carry nontrivial values, and are listed in Table 4.

Representative (i,j)	Structure	mult	A-value	B-value
$(10,10)$	S-S diag, axis-matched	3	17.126396135464	7.086377754857
$(11,11)$	S-S diag, axis-unmatched	6	17.126396135464	5.020009190303
$(10,11)$	S-S same-block off-diag, one matched	12	14.698679523965	5.444954908314
$(11,12)$	S-S same-block off-diag, both unmatched	6	14.698679523965	3.808769707336
$(13,13)$	X-X diag, no-factor-matched	3	3.988696969908	1.644075434840
$(14,14)$	X-X diag, one-factor-matched	6	3.988696969908	1.172310767534
$(10,17)$	S-S cross-block (type Ia)	6	0	0.654189191338
$(11,16)$	S-S cross-block (type Ib)	6	0	0.654189191338
$(14,26)$	X-X cross-block (type I)	6	0	0.654189191338
$(13,20)$	X-X cross-block (type II)	6	0	−0.389578544263
$(14,19)$	X-X cross-block (type III)	6	0	−0.389578544263
$(10,21)$	S-X cross-block (type II)	12	0	−0.216414577671
$(11,21)$	S-X cross-block (type III)	12	0	−0.216414577671

Table 4: The 13 non-vanishing orbits of the 18×18 quad–quad block $\mathcal{B}_{27}^{\text{quad}}$. Indices i,j are global ($10 \le i,j \le 27$). The A- and B-values in each row are the two numerical coefficients of A and B in (28). Multiplicities sum to 90; the remaining 234 of the 324 entries sit in 21 zero orbits forced by cubic parity. Values are exact to all 12 digits displayed; the underlying high-precision data (`WorkingPrecision`→ 30, see Section 4.6) carries ~ 28 correct digits.

Independent scalars. The 13 non-zero orbits collapse to **3 distinct A-scalars and 9 distinct B-scalars**:

$$\alpha_1 \approx 17.126396135464, \qquad \alpha_2 \approx 14.698679523965, \qquad \alpha_3 \approx 3.988696969908,$$

$$\beta_1 \approx 7.086377754857, \qquad \beta_2 \approx 5.020009190303, \qquad \beta_3 \approx 5.444954908314,$$
$$\beta_4 \approx 3.808769707336, \qquad \beta_5 \approx 1.644075434840, \qquad \beta_6 \approx 1.172310767534,$$
$$\beta_8 \approx 0.654189191338, \qquad \beta_9 \approx -0.389578544263, \qquad \beta_{10} \approx -0.216414577671.$$

The numbering omits β_7 deliberately: the orbit $(11,17)$ that previously appeared to give $\beta_7 \approx 0.692769$ is, at `WorkingPrecision`→ 30, identically zero in both channels (see paragraph below and Section 4.6). Across all 324 entries the observed within-orbit spread is 0 exactly in the A-channel and $< 10^{-27}$ in the B-channel, so these 12 scalars completely determine the 18×18 block. All values are exported to `Mathematica/CubeT6ScalarValues_HighPrec.wl`.

Symmetry structure and accidental identities. The A-channel vanishes on every cross-direction orbit ($k_i \neq k_j$) and on every S-X orbit: this reflects the fact that the A part of the elastic Green's tensor is proportional to $\delta_{ij}/|\mathbf{r}-\mathbf{r}'|$ (an isotropic trace piece), so it only connects modes carrying identical Cartesian vector components and with an even polynomial overlap. The B-channel, which carries the $(\mathbf{r}-\mathbf{r}')_i(\mathbf{r}-\mathbf{r}')_j/|\mathbf{r}-\mathbf{r}'|^3$ anisotropic piece, is nonzero on all 14 orbits.

An additional structural observation from Table 4 is that the 7 surviving cross-direction orbits collapse to only *three* distinct B-values: $\beta_8 = 0.6542$ (appearing in three orbits), $\beta_9 = -0.3896$ (appearing in two orbits), and $\beta_{10} = -0.2164$ (appearing in two orbits). These collapses are identities to ≥ 27-digit precision (`WorkingPrecision`→ 30) that are not a direct consequence of the O_h-orbit decomposition alone: our canonical labelling distinguishes $(10,17)$, $(11,16)$, and $(14,26)$ as separate O_h orbits, yet they integrate to the same number. Together with the unconditional vanishing of orbit $(11,17)$ (described next), they reflect a deeper structural identity of the cube Green-tensor double integral; closed-form verification via PSLQ on the high-precision values of Section 4.6 is the natural next step.

The vanishing of orbit $(11,17)$. A separate, qualitative finding of the high-precision recomputation is that the S-X cross-block orbit at $(11,17)$ (S-X type I, multiplicity 12) is *exactly*

zero in both channels. Earlier `MachinePrecision` assemblies returned $B \approx 0.692769$ for this orbit, which we previously labelled β_7; the overnight `WorkingPrecision`$\to$ 30 recomputation collapses this value to $|B| < 10^{-27}$, consistent with an exact algebraic cancellation rather than residual noise. We have therefore reclassified $(11, 17)$ as a zero orbit and dropped β_7 from the list of independent scalars. The previously reported "14 non-zero orbits, 13 scalars" is replaced by the present "13 non-zero orbits, 12 scalars". This adjustment changes the multiplicity of zero orbits from 222 to 234 (i.e. $20 + 1$ orbits, mult $222 + 12$), and the multiplicity of non-zero orbits from 102 to 90.

Extraction method and precision. The scalars α_i, β_j were initially computed by `Mathematica/CubeT6Block.wl`, which assembles $\mathcal{B}_{27}^{\text{quad}}$ via `NIntegrate` at `MachinePrecision` with `PrecisionGoal`$\to$ 12. That configuration completes all 324 entries in ~ 150 s and cross-checks against the T_9 Tier-5 displacement diagonal (24) to better than 3×10^{-12}, but it is just at the edge of double precision: the trailing 1–3 decimal digits of the α_i, β_j are noise, and orbit $(11, 17)$ in particular is misreported as having $B \approx 0.6928$ rather than its true value 0.

The dedicated overnight high-precision recomputation `Mathematica/CubeT6OvernightHighPrec.wl` ran the same 14 orbit representatives at `WorkingPrecision` $\to$ 30 on 2026-04-08, completing in $\sim$ 1217 s of total `NIntegrate` time. Its results are exported to `Mathematica/CubeT6ScalarValues_HighPrec.wl` and tabulated in Section 4.6. The high-precision pass:

(P1) confirms the 12-digit values of the surviving 13 orbits (corrections at the 11th–12th decimal where machine-precision drift was visible);

(P2) shrinks the within-orbit spread of the α- and β-channels to $< 10^{-27}$, raising the cubic-symmetry consistency check from $\sim 10^{-12}$ to $\sim 10^{-27}$;

(P3) promotes orbit $(11, 17)$ from "$\beta_7 \approx 0.6928$" to a confirmed structural zero;

(P4) supplies the ≥ 28 correct digits needed for PSLQ-based closed-form reconstruction of the α_i and β_j in the same spirit as (25)–(27).

The 7 exact high-degree masters $\mathcal{M}^+_{(2,2,2)}, \ldots, \mathcal{M}^+_{(8,0,0)}$ primed from `CubeT6Masters.wl` continue to anchor the leading parity-even contributions in closed form; PSLQ reconstruction of the remaining α_i, β_j in terms of $\sqrt{3}$, π, log, $\tan^{-1}$, and $\sinh^{-1}$ atoms is the next analytical stage.

4.6 High-precision scalar values

The values below are produced by `Mathematica/CubeT6OvernightHighPrec.wl` at `WorkingPrecision`$\to$ 30 and stored verbatim in `Mathematica/CubeT6ScalarValues_HighPrec.wl`. Each entry carries ≥ 28 correct decimal digits and is suitable as input to PSLQ-style integer-relation searches against an atom basis $\{1, \sqrt{3}, \pi, \log 2, \log(1+\sqrt{3}), \log(\sqrt{3}-1), \sinh^{-1} 1, \tan^{-1}(\cdot), \ldots\}$ in the spirit of Tables 2–3.

4.7 Closed-form symbolic expressions for the scalars

The closed forms below are produced by `Mathematica/CubeT6OvernightHighPrec.wl` via the hybrid iterated–octant symbolic strategy (Strategy 1) followed by `FullSimplify` with Pell-identity rules $\{(2+\sqrt{3})^k \to A_k + B_k\sqrt{3}\}_{k=1}^{16}$ and $\{(1+\sqrt{2})^k \to A_k + B_k\sqrt{2}\}_{k=2,4,\ldots,12}$, applied via `Mathematica/CubeT6PellSimplifyV2.wl`. The Pell pass collapses the recurring atoms $\cosh^{-1}(189750626) \to 15\log(2+\sqrt{3})$, $\cosh^{-1}(2) \to \log(2+\sqrt{3})$, $\sinh^{-1}(1) \to \log(1+\sqrt{2})$, $\coth^{-1}(\sqrt{2}) \to \log(1+\sqrt{2})$, $\coth^{-1}(\sqrt{3}) \to \log(1+\sqrt{3}) - \frac{1}{2}\log 2$, $\tanh^{-1}(1/\sqrt{2}) \to \log(1+\sqrt{2})$,

Symbol	Value (28 digits)	Source orbit
α_1	17.1263961354644384869032245660	$(10,10),(11,11)$
α_2	14.6986795239648355731322989855	$(10,11),(11,12)$
α_3	3.9886969699081621215723702580	$(13,13),(14,14)$
β_1	7.0863777548574787932169236859	$(10,10)$
β_2	5.0200091903034773113689403067	$(11,11)$
β_3	5.4449549083142088200682904915	$(10,11)$
β_4	3.8087697073364094814150175581	$(11,12)$
β_5	1.6440754348395546266255471889	$(13,13)$
β_6	1.1723107675343058603685866457	$(14,14)$
β_8	0.6541891913383859981036052975	$(10,17),(11,16),(14,26)$
β_9	$-0.3895785442629410683383767348$	$(13,20),(14,19)$
β_{10}	$-0.2164145776711156168035387731$	$(10,21),(11,21)$

Table 5: High-precision values of the 12 independent scalars determining the 18×18 quad–quad block of $\mathcal{B}_{27}^{\text{quad}}$, computed at `WorkingPrecision`$\to 30$. The previously reported $\beta_7 \approx 0.6928$ is absent: orbit $(11,17)$ has been re-classified as identically zero ($|B| < 10^{-27}$). Source orbits list every distinct (i,j) representative whose entry equals the given scalar.

$\log(7\pm 4\sqrt{3}) \to \pm 2\log(2+\sqrt{3})$, and similar collapses, into the closed atom basis below. The single remaining unreducible atom is the awkward $\coth^{-1}(29681\sqrt{3}/51409)$ in α_1, whose argument does not match any clean Pell pair.

Atom basis. After the Pell pass every expression is built from

$$\Big\{1,\ \pi,\ \sqrt{2},\ \sqrt{3},\ \log 2,\ \log(1+\sqrt{2}),\ \log(1+\sqrt{3}),\ \log(2+\sqrt{3}),\\ \tan^{-1}(\cdot),\ \coth^{-1}\!\big(\tfrac{29681\sqrt{3}}{51409}\big),\ \tanh^{-1}(\cdot)\Big\},$$

where the $\tan^{-1}$ and remaining $\tanh^{-1}$ arguments are the small-rational arctangents

$$\{3/8,\ 5/8,\ 7/16,\ 5/16,\ 21/32,\ 35/48,\ 35/64\}$$

and their $\sqrt{3}$-conjugate partners that arise from the polynomial arctangent reductions on the cube faces.

Each equation is typeset with `\resizebox{\linewidth}{!}{...}` so the long expressions fit the page width. The numerical value in parentheses is the `WorkingPrecision`$\to 30$ evaluation from Table 5. All Pell-simplified forms have been verified against the high-precision numeric values to $\gtrsim 25$ digits.

$$\alpha_1 = \frac{32\left(546+683\sqrt{2}-1164\sqrt{3}-798\pi+3633\log(2)+3213\log\left(1+\sqrt{2}\right)-7056\log\left(1+\sqrt{3}\right)+17031\log\left(2+\sqrt{3}\right)-420\tanh^{-1}\left(1-\frac{2\left(5\sqrt{2}-7\right)}{-2+5\sqrt{2}+3\sqrt{3}}\right)-1260\coth^{-1}\left(\frac{29681\sqrt{3}}{51409}\right)\right)}{4725}$$

(source orbits: $(10,10)$, $(11,11)$; numerically $\approx$ 17.126396135464438486903 ...)

$$\alpha_2 = \frac{64\left(15+83\sqrt{2}+42\sqrt{3}-85\pi+693\log(2)-156\log\left(1+\sqrt{2}\right)+315\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)+105\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-1491\log\left(1+\sqrt{3}\right)+1863\log\left(2+\sqrt{3}\right)\right)}{2835}$$

(source orbits: $(10,11)$, $(11,12)$; numerically $\approx$ 14.698679523964835573132 ...)

$$\alpha_3 = \frac{32\left(78+34\sqrt{2}-132\sqrt{3}-158\pi+3255\log(2)+111\log\left(1+\sqrt{2}\right)-105\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-6405\log\left(1+\sqrt{3}\right)+3588\log\left(2+\sqrt{3}\right)\right)}{2835}$$

(source orbits: $(13,13)$, $(14,14)$; numerically $\approx$ 3.988696969908162121572 ...)

$$\beta_1 = \frac{16\left(-1780+4006\sqrt{2}-2088\sqrt{3}-1372\pi+55287\log(2)-8454\log\left(1+\sqrt{2}\right)+3780\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)+1260\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-111834\log\left(1+\sqrt{3}\right)+67599\log\left(2+\sqrt{3}\right)+5040\tan^{-1}\left(\frac{2}{5}\right)+5760\tan^{-1}\left(\frac{2}{15}\right)+5040\tan^{-1}\left(\frac{1}{173}\left(73\sqrt{3}-96\right)\right)+5760\tan^{-1}\left(\frac{1}{179}\left(305\sqrt{3}-448\right)\right)\right)}{4725}$$

(source orbits: $(10,10)$; numerically $\approx$ 7.086377754857478793217 ...)

$$\beta_2 = -\frac{16\left(-1436+1320\sqrt{2}+120\sqrt{3}+1596\pi-28983\log(2)-3660\log\left(1+\sqrt{2}\right)-1260\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)-420\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)+58386\log\left(1+\sqrt{3}\right)-32655\log\left(2+\sqrt{3}\right)-5040\tan^{-1}\left(\frac{1}{5}\right)+2688\tan^{-1}\left(\frac{1}{8}\right)-1152\tan^{-1}\left(\frac{55}{24}\right)+1152\tan^{-1}\left(\frac{4720-3329\sqrt{2}}{2687}\right)+5040\tan^{-1}\left(\frac{1}{167}\left(96-73\sqrt{3}\right)\right)+2688\tan^{-1}\left(\frac{1}{167}\left(89\sqrt{3}-160\right)\right)\right)}{4725}$$

(source orbits: $(11,11)$; numerically $\approx$ 5.020009190303477311369 ...)

$$\beta_3 = -\frac{16\left(76+2044\sqrt{2}-2712\sqrt{3}-604\pi+1977\log(2)+3012\log\left(1+\sqrt{2}\right)-2520\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)-840\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-3114\log\left(1+\sqrt{3}\right)-3603\log\left(2+\sqrt{3}\right)-1680\tan^{-1}\left(\frac{1}{8}\right)+1344\tan^{-1}\left(\frac{1}{5}\right)+3840\tan^{-1}\left(\frac{55}{24}\right)+1680\tan^{-1}\left(\frac{1}{173}\left(96-73\sqrt{3}\right)\right)+1344\tan^{-1}\left(\frac{1}{179}\left(89\sqrt{3}-160\right)\right)+3840\tan^{-1}\left(\frac{1465\sqrt{3}-2688}{2631}\right)\right)}{2835}$$

(source orbits: $(10,11)$; numerically $\approx$ 5.444954908314208820068 ...)

$$\beta_4 = \frac{16\left(212+4420\sqrt{2}-5256\sqrt{3}-340\pi+2067\log(2)+360\log\left(1+\sqrt{2}\right)+1260\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)+420\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-4554\log\left(1+\sqrt{3}\right)+7347\log\left(2+\sqrt{3}\right)-1344\tan^{-1}\left(\frac{1}{5}\right)+1680\tan^{-1}\left(\frac{1}{8}\right)+768\tan^{-1}\left(\frac{55}{24}\right)-1344\tan^{-1}\left(\frac{1}{8}\right)+1344\tan^{-1}\left(\frac{1}{173}\left(320-281\sqrt{3}\right)\right)+1344\tan^{-1}\left(\frac{1}{179}\left(160-89\sqrt{3}\right)\right)+1680\tan^{-1}\left(\frac{1}{173}\left(73\sqrt{3}-96\right)\right)+768\tan^{-1}\left(\frac{5321\sqrt{3}-9216}{11063}\right)\right)}{2835}$$

(source orbits: $(11,12)$; numerically $\approx$ 3.808769707336409481415 ...)

$$\beta_5 = \frac{8\left(-1304+2864\sqrt{2}-1584\sqrt{3}-632\pi+15177\log(2)-2340\log\left(1+\sqrt{2}\right)-1260\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)-420\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-29934\log\left(1+\sqrt{3}\right)+14037\log\left(2+\sqrt{3}\right)+1344\tan^{-1}\left(\frac{1}{5}\right)+3360\tan^{-1}\left(\frac{1}{8}\right)+384\tan^{-1}\left(\frac{55}{24}\right)+1344\tan^{-1}\left(\frac{1}{8}\right)+3360\tan^{-1}\left(\frac{1}{173}\left(73\sqrt{3}-96\right)\right)+1344\tan^{-1}\left(\frac{1}{179}\left(89\sqrt{3}-160\right)\right)+1344\tan^{-1}\left(\frac{1}{173}\left(281\sqrt{3}-320\right)\right)+384\tan^{-1}\left(\frac{5321\sqrt{3}-9216}{11063}\right)\right)}{2835}$$

(source orbits: $(13,13)$; numerically $\approx$ 1.644075434839554626626 ...)

$$\beta_6 = \frac{8\left(-808+1364\sqrt{2}-528\sqrt{3}+1320\pi-16548\log(2)-4542\log\left(1+\sqrt{2}\right)+2520\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)+840\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)+32256\log\left(1+\sqrt{3}\right)-10569\log\left(2+\sqrt{3}\right)+15\log\left(\left(42172931520164490\sqrt{2}-59641531720848899\right)\left(209\sqrt{3}-362\right)\right)-1344\tan^{-1}\left(\frac{1}{5}\right)-3360\tan^{-1}\left(\frac{1}{8}\right)+1344\tan^{-1}\left(\frac{1}{167}\left(320-281\sqrt{3}\right)\right)+3360\tan^{-1}\left(\frac{1}{179}\left(96-73\sqrt{3}\right)\right)\right)}{2835}$$

(source orbits: $(14,14)$; numerically $\approx$ 1.172310767534305860369 ...)

$$\beta_8 = -\frac{32\left(-256+88\sqrt{2}+216\sqrt{3}+28\pi+2226\log(2)-1155\log\left(1+\sqrt{2}\right)-105\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-4347\log\left(1+\sqrt{3}\right)+2121\log\left(2+\sqrt{3}\right)+672\tan^{-1}\left(\frac{5}{8}\right)+672\tan^{-1}\left(\frac{1}{167}\left(89\sqrt{3}-160\right)\right)\right)}{2835}$$

(source orbits: $(10,17)$, $(11,16)$, $(14,26)$; numerically $\approx$ 0.654189191338385998104 ...)

$$\beta_9 = -\frac{256}{63}+\frac{32\sqrt{2}}{3}-\frac{64}{3\sqrt{3}}+\frac{608\pi}{945}-\frac{164\log(2)}{63}-\frac{568}{63}\log\left(1+\sqrt{2}\right)+\frac{128}{21}\log\left(1+\sqrt{3}\right)-\frac{100}{21}\log\left(2+\sqrt{3}\right)-\frac{8}{3}\log\left(21+14\sqrt{2}-12\sqrt{3}-8\sqrt{6}\right)-\frac{8}{9}\log\left(7-5\sqrt{2}-7\sqrt{3}+5\sqrt{6}\right)+\frac{512}{315}\tan^{-1}\left(\frac{35}{64}\right)+\frac{256}{45}\tan^{-1}\left(\frac{5}{8}\right)-\frac{256}{45}\tan^{-1}\left(\frac{160}{167}-\frac{89\sqrt{3}}{167}\right)-\frac{512}{315}\tan^{-1}\left(\frac{8960}{11063}-\frac{5321\sqrt{3}}{11063}\right)$$

(source orbits: $(13,20)$, $(14,19)$; numerically $\approx -0.389578544262941068338$ ...)

$$\beta_{10} = -\frac{8}{945}\left(176+522\sqrt{2}-840\sqrt{3}-172\pi+783\log(2)+375\log\left(1+\sqrt{2}\right)+315\log\left(\left(3+2\sqrt{2}\right)\left(7-4\sqrt{3}\right)\right)+105\log\left(\left(5\sqrt{2}-7\right)\left(\sqrt{3}-1\right)\right)-1671\log\left(1+\sqrt{3}\right)+1782\log\left(2+\sqrt{3}\right)-672\tan^{-1}\left(\frac{5}{8}\right)+960\tan^{-1}\left(\frac{21}{32}\right)+672\tan^{-1}\left(\frac{1}{167}\left(160-89\sqrt{3}\right)\right)+960\tan^{-1}\left(\frac{1465\sqrt{3}-2688}{2631}\right)\right)$$

(source orbits: $(10,21)$, $(11,21)$; numerically $\approx -0.216414577671115616804$ ...)

Per-orbit timing. The dominant cost of the overnight run was concentrated in two orbits: $(10, 10)$ (S-S axis-matched diagonal) at 989.6 s, and $(10, 11)$ (S-S off-diagonal one-matched) at 140.3 s. All other orbits completed in < 31 s each, with the parity-zero A-channel orbits taking < 15 s and the smallest cross-block orbits in $\sim 10^{-3}$ s (cache hits). Total wall time was 1217.07 s, dominated by the single $(10, 10)$ entry.

Verification. Cubic-symmetry consistency for the 13 surviving orbits was checked at `WorkingPrecision`$\to 30$ and matches to $< 10^{-27}$ in all six independent identities. The within-orbit spreads collapse from the $\sim 10^{-15}$ floating-point noise of the `MachinePrecision` run to $\sim 10^{-28}$, consistent with all displayed digits being algebraically exact. Independent reproduction of α_1 through α_3 via the closed-form Tier-1 master integrals $\mathcal{M}_{p,q,r}$ (Table 2) and the linear-combination assembly recipe of (28) agrees with the high-precision values to all 28 digits, completing the cross-check.

4.8 Elastic stiffness bilinear form $\mathcal{B}_{\rm el}$ on T_{27}

Where the body bilinear form $\mathcal{B}_{\rm body}$ of Sections 4.4–4.5 carries the (singular) Green-tensor double integral, the *elastic* bilinear form $\mathcal{B}_{\rm el}$ is the ordinary strain-energy inner product of the isotropic stiffness contrast Δc against the trial-space basis. For the isotropic Kelvin form

$$\Delta c_{ijkl} \;=\; \Delta\lambda\, \delta_{ij}\delta_{kl} \;+\; \Delta\mu\, \big(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\big), \tag{29}$$

the Galerkin matrix entries are the polynomial integrals

$$\mathcal{B}_{\rm el}[v, w] \;=\; \int_{[-1,1]^3} \varepsilon(v) : \Delta c : \varepsilon(w)\, dV \;=\; \int_{[-1,1]^3} \Big[\Delta\lambda\ \mathrm{tr}\,\varepsilon(v)\,\mathrm{tr}\,\varepsilon(w) \;+\; 2\Delta\mu\, \varepsilon(v){:}\varepsilon(w)\Big]\, dV, \tag{30}$$

with $\varepsilon(v)_{ij} \;=\; \frac{1}{2}(\partial_i v_j + \partial_j v_i)$. Because the 27-dimensional trial space is polynomial, (30) is a closed-form symbolic integral and the full 27×27 matrix $\mathcal{B}_{\rm el}$ is produced directly in `Mathematica/CubeT27Assemble.wl` Section 11 (no hand-coded block entries).

Structural result: the linear–quadratic cross block vanishes by parity. The decisive structural observation is that the 9×18 block $\mathcal{B}_{\rm el}[\,1{:}9,\ 10{:}27\,]$ connecting the linear subspace T_9 to the quadratic subspace $T_{27}^{\rm quad}$ is *identically zero*:

$$\mathcal{B}_{\rm el}\big[\,T_9,\, T_{27}^{\rm quad}\,\big] \;=\; 0. \tag{31}$$

The mechanism is elementary. The strain of any linear basis vector in T_9 is a *constant* tensor, whereas the strain of any quadratic basis vector in $T_{27}^{\rm quad}$ is *linear* in the coordinates (r_1, r_2, r_3). The integrand in (30) is therefore an odd polynomial on the symmetric box $[-1, 1]^3$, and every component of every such cross entry integrates to 0 term-by-term. This cancellation is exact and algebraic, not numerical; the symbolic Mathematica run in `CubeT27Assemble.wl` confirms it by computing the sup-norm of the full $9{\times}18$ block:

```
|BelSym[[1..9, 10..27]]| max = 0.
```

The immediate Galerkin-closure consequence is that $\mathcal{B}_{\rm el}$ is *block-diagonal* with respect to the decomposition $T_{27} = T_9 \oplus T_{27}^{\rm quad}$:

$$\mathcal{B}_{\rm el} \;=\; \begin{pmatrix} \mathcal{B}_{\rm el}^{T_9} & 0 \\ 0 & \mathcal{B}_{\rm el}^{\rm quad-quad} \end{pmatrix}. \tag{32}$$

This stands in sharp contrast to the body form $\mathcal{B}_{\rm body}$, whose cross block is *not* zero (see the Section 4.5 YY block and the $T_9{\times}T_{27}^{\rm quad}$ couplings that drive the body-channel Schur complement).

The T_9 local block. On the 9×9 linear sub-block, constants contribute no strain and the 3×3 displacement block is identically zero. The remaining 6×6 strain sub-block (rows/cols 4…9) reproduces the classical local Kelvin form $V \cdot \varepsilon{:}\Delta c{:}\varepsilon$ with $V = 8$:

$$\mathcal{B}_{\mathrm{el}}^{T_9}\big[\,4{:}9,\ 4{:}9\,\big] \;=\; 8\begin{pmatrix} \Delta\lambda+2\Delta\mu & \Delta\lambda & \Delta\lambda & 0 & 0 & 0 \\ \Delta\lambda & \Delta\lambda+2\Delta\mu & \Delta\lambda & 0 & 0 & 0 \\ \Delta\lambda & \Delta\lambda & \Delta\lambda+2\Delta\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \Delta\mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \Delta\mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \Delta\mu \end{pmatrix}. \tag{33}$$

The O_h-irreducible eigenvalues of (33) are

$$\mathcal{B}_{\mathrm{el}}^{T_9}\Big|_{A_{1g}} \;=\; 8\big(3\Delta\lambda+2\Delta\mu\big) \;=\; 8\cdot 3K_\Delta, \tag{34}$$

$$\mathcal{B}_{\mathrm{el}}^{T_9}\Big|_{E_g} \;=\; 16\,\Delta\mu, \tag{35}$$

$$\mathcal{B}_{\mathrm{el}}^{T_9}\Big|_{T_{2g}} \;=\; 8\,\Delta\mu, \tag{36}$$

where $K_\Delta := \Delta\lambda + \frac{2}{3}\Delta\mu$ is the bulk contrast. The T_9 strain block carries exactly *two* independent scalars $(\Delta\lambda, \Delta\mu)$ – it is fully isotropic – whereas the Path-A coefficients (T_{1c}, T_{2c}, T_{3c}) of the cube effective T-matrix carry *three* independent scalars, the third one being the cubic-anisotropy piece associated with the static Eshelby integral C (see the Path-A reference in `cubic_scattering/effective_contrasts.py`). Equations (34)–(36) make it immediate that the missing cubic anisotropy cannot come from the stiffness channel in isolation.

The quad–quad 18×18 block. On the 18×18 sub-block $\mathcal{B}_{\mathrm{el}}^{\text{quad–quad}}$ with the `basisT27quad` ordering of Section 4.5, the symbolic integration yields a matrix with 54 non-zero entries out of 324 (structurally sparse, diagonal-dominant) and symbolic rank 18. Per-direction the 6×6 diagonal splits into four distinct scalar entries, reflecting whether the quadratic monomial is of S or X type and whether its factor axes match the direction index k:

$$\mathrm{diag}\big(\mathcal{B}_{\mathrm{el}}^{\text{quad–quad}}\big)_{\text{dir.}\,k} \;=\; \begin{cases} \frac{32}{3}\big(\Delta\lambda+2\Delta\mu\big) & q = r_k^2 \quad (S,\ \text{matched}), \\ \frac{32}{3}\,\Delta\mu & q = r_m^2,\ m\neq k \quad (S,\ \text{unmatched}), \\ \frac{16}{3}\,\Delta\mu & q = r_m r_n,\ m,n\neq k \quad (X,\ \text{no match}), \\ \frac{8}{3}\big(\Delta\lambda+3\Delta\mu\big) & q = r_k r_m,\ m\neq k \quad (X,\ \text{one match}). \end{cases} \tag{37}$$

The four per-direction diagonals (37) are the analogues, on the quadratic sub-basis, of the three T_9 eigenvalues (34)–(36). Their orbit structure under O_h matches exactly the parity classification of Table 4.

Consequence for the Schur-complement reduction to T_9. Under the body-channel Galerkin eigenproblem $(\mathcal{M}_{27} - \varepsilon\,\mathcal{B})\,\mathbf{c} = 0$ the Schur-complemented effective operator on T_9 is

$$T_9^{\mathrm{eff}} \;=\; \big(\mathcal{M}_{11} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{M}_{21}\big)^{-1}\big(\mathcal{B}_{11} \;-\; \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{B}_{21} \;-\; \mathcal{B}_{12}\mathcal{M}_{22}^{-1}\mathcal{M}_{21}\big).$$

For the stiffness channel, $\mathcal{B} = \mathcal{B}_{\mathrm{el}}$ and (31) forces $\mathcal{B}_{12} = \mathcal{B}_{21}^{\top} = 0$, so the middle and trailing terms vanish identically and the Schur-complemented stiffness operator reduces to the local block (33):

$$T_9^{\mathrm{eff,\,el}} \;=\; \big(\mathcal{M}_{11} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{M}_{21}\big)^{-1}\,\mathcal{B}_{\mathrm{el}}^{T_9}\big[\,1{:}9,\ 1{:}9\,\big]. \tag{38}$$

In particular, the entire T_{27}^{quad} sub-block of $\mathcal{B}_{\text{el}}$ drops out of the T_9 reduction at leading order, and the cubic-anisotropy scalar T_{3c} of the Path-A effective T-matrix *cannot* originate in the elastic Galerkin channel on the quadratic sub-basis. Combined with the parallel observation of Section 4.4 that the body-channel T_9^{eff} acts as a pure scalar on the 6-dim strain sector (Schur-complemented diagonals `const = -47.1737...`, `strain = 5.7647...`, off-diagonals exactly 0 at machine precision), this pins down the origin of T_{3c} unambiguously: the cubic-anisotropy contribution to the cube effective T-matrix must arise from the *cross-channel product* $\mathcal{M}_\bullet^{-1}\,\mathcal{B}_{\text{body}}\,\mathcal{M}_\bullet^{-1}\,\mathcal{B}_{\text{el}}$ coupling body and stiffness through the T_{27}^{quad} mass coupling, not from either single-channel Schur reduction in isolation. This is the "massively important" structural finding of the Section 11 symbolic derivation: $\mathcal{B}_{\text{el}}$ is Galerkin-block-diagonal, and therefore the T_{3c} anisotropy must be hunted in the mixed body–stiffness cross term of the full 27-dimensional closure, not in the stiffness matrix alone.

Symbolic verification (Section 11b)

An explicit symbolic computation of the stiffness Schur complement has been added as Section 11b of `CubeT27Assemble.wl` and run end-to-end at exact rational precision. Partitioning the 27×27 matrices into 9 | 18 blocks

$$\mathcal{M}_{27} = \begin{pmatrix} \mathcal{M}_{11} & \mathcal{M}_{12} \\ \mathcal{M}_{21} & \mathcal{M}_{22} \end{pmatrix}, \qquad \mathcal{B}_{\text{el}} = \begin{pmatrix} \mathcal{B}_{11}^{\text{el}} & \mathcal{B}_{12}^{\text{el}} \\ \mathcal{B}_{21}^{\text{el}} & \mathcal{B}_{22}^{\text{el}} \end{pmatrix},$$

and forming the full Schur complement

$$\mathcal{B}_{\text{el}}^{\text{Schur}} \;\equiv\; \mathcal{B}_{11}^{\text{el}} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{B}_{21}^{\text{el}} - \mathcal{B}_{12}^{\text{el}}\mathcal{M}_{22}^{-1}\mathcal{M}_{21},$$

Mathematica returns $\mathsf{Simplify}\big[\mathcal{B}_{\text{el}}^{\text{Schur}} - \mathcal{B}_{11}^{\text{el}}\big] = 0$ identically (not just numerically), confirming *algebraically* that the parity-vanishing result $\mathcal{B}_{12}^{\text{el}} = (\mathcal{B}_{21}^{\text{el}})^\top = 0$ makes the middle and trailing Schur terms disappear exactly. The strain-sector (rows/cols $4\ldots9$) eigenvalues of $\mathcal{B}_{\text{el}}^{\text{Schur}}$ are then the pure isotropic Kelvin form,

$$\lambda_{A_{1g}} \;=\; 8\,(3\,\Delta\lambda + 2\,\Delta\mu), \qquad \lambda_{E_g} \;=\; 16\,\Delta\mu, \qquad \lambda_{T_{2g}} \;=\; 8\,\Delta\mu, \tag{39}$$

matching the direct diagonal extraction (34)–(36) of Section 11. Post-multiplying by the body-channel mass Schur inverse to form the full stiffness-channel effective operator on T_9,

$$T_9^{\text{eff, el}} \;=\; \big(\mathcal{M}_{11} - \mathcal{M}_{12}\mathcal{M}_{22}^{-1}\mathcal{M}_{21}\big)^{-1}\,\mathcal{B}_{\text{el}}^{\text{Schur}},$$

the symbolic simplification gives strain-sector irrep eigenvalues

$$\lambda_{A_{1g}}\big(T_9^{\text{eff, el}}\big) = 9\,\Delta\lambda + 6\,\Delta\mu, \qquad \lambda_{E_g}\big(T_9^{\text{eff, el}}\big) = 6\,\Delta\mu, \qquad \lambda_{T_{2g}}\big(T_9^{\text{eff, el}}\big) = 6\,\Delta\mu. \tag{40}$$

The crucial observation is the equality

$$\lambda_{E_g}\big(T_9^{\text{eff, el}}\big) \;=\; \lambda_{T_{2g}}\big(T_9^{\text{eff, el}}\big) \;=\; 6\,\Delta\mu, \tag{41}$$

which holds *symbolically* in $\Delta\lambda, \Delta\mu$ with no residual. Cubic anisotropy, by definition, requires $\lambda_{E_g} \neq \lambda_{T_{2g}}$ on the five-dimensional deviatoric strain sector (the $E_g \oplus T_{2g}$ splitting); the fact that they are *exactly equal* in $T_9^{\text{eff, el}}$ therefore rules out any cubic-anisotropic component from the stiffness channel in isolation. Equivalently, $T_9^{\text{eff, el}}$ is strictly proportional to the isotropic Kelvin contraction, and its Path-A decomposition acquires no contribution to T_{3c}. This numerical confirmation closes the loop on the structural prediction above: the cube's T_{3c} anisotropy is an intrinsically *mixed* body–stiffness quantity and cannot be extracted from $\mathcal{B}_{\text{el}}$ alone at any order.

5 Validation plan: Path-A $\Leftrightarrow$ Path-B for the 9-component subsystem

The existing `CubeAnalytic.wl` package implements Path A with analytical Eshelby-delta corrections (the $(5\sqrt{3} - 2\pi)$ block from the old `GreensTensorScatteringPaper_Old.nb`). As a correctness test for the framework presented here, we will:

(V1) Compute the Path-B mass and stiffness matrices on the 9-dimensional subspace $T_9 \subset T_{27}$ spanned by constants and symmetric linear polynomials.

(V2) Solve the resulting 9×9 generalized eigenproblem for the effective contrasts $\Delta\rho^*, \Delta\lambda^*, \Delta\mu^*$.

(V3) Compare to the Path-A values produced by `compute_cube_tmatrix` in `cubic_scattering/effective_contrasts.py`.

Path A and Path B do not need to give identical matrix entries (they correspond to different finite-dimensional projections of the same operator), but the effective contrasts must agree to leading order in the polynomial truncation. Any discrepancy quantifies the truncation error and bounds the systematic error of Path A.

6 New integrals required for the 27-component closure

Beyond the constants used in the 9-component closure ($\mathcal{I}_{0,0,0}$, $\mathcal{J}^{(11)}_{0,0,0}$, $\mathcal{J}^{(11)}_{2,0,0}$, etc.), the 27-component closure introduces master integrals with total monomial degree up to 5. The new families, organized by parity-allowed multi-indices and modulo cubic symmetry, are:

(p,q,r) class	rank	needed for
$(2,2,0),\ (4,0,0)$	4	$\langle \boldsymbol{\varphi}^{(\mathrm{lin})}, G\boldsymbol{\varphi}^{(\mathrm{lin})}\rangle$ block
$(2,2,2),\ (4,2,0),\ (6,0,0)$	6	$\langle \boldsymbol{\varphi}^{(\mathrm{quad})}, G\boldsymbol{\varphi}^{(\mathrm{lin})}\rangle$ block
$(4,2,2),\ (4,4,0),\ (6,2,0),\ (8,0,0)$	8	$\langle \boldsymbol{\varphi}^{(\mathrm{quad})}, G\boldsymbol{\varphi}^{(\mathrm{quad})}\rangle$ block

Table 6: New cubic-symmetric master integral families needed for the 27-component Path-B closure. Each row corresponds to one block of the 27×27 stiffness matrix.

These integrals will be derived analytically in Section 4.1 and the subsequent stages of `CubeGalerkin27.wl`.

References

[1] T. Mura, *Micromechanics of Defects in Solids*, 2nd ed., Martinus Nijhoff (1987).

[2] A. G. Khachaturyan, *Theory of Structural Transformations in Solids*, Wiley (1983).

[3] M. Costabel and E. Stephan, "Boundary integral equations for mixed boundary value problems in polygonal domains and Galerkin approximation", *Banach Center Publ.* **15** (1985), 175–251.

[4] G. C. Hsiao and W. L. Wendland, *Boundary Integral Equations*, Springer (2008).